of quaternions, he later recalled. In a less poetic metaphor, he compared his satisfaction upon solving the problem to "an intellectual want relieved" [**17**, 2:435–436]. Hamilton immediately pulled out his notebook and jotted down the fundamental quaternion formula. Anxious to check the consistency of his new algebra, he continued scribbling in a carriage on the way to a meeting of the Council of the Royal Irish Academy and later while presiding over that meeting as Academy President. Neither chatting with his wife nor chairing an academic meeting apparently interfered with the train of his mathematical thought.

Hamilton felt that he might explore the ramifications of the quaternion theory for 10–15 years — as long as he had spent trying to work it out. He speculated that the scalar part of the quaternion might express quantity (say, of electricity), while the imaginary parts might determine direction and intensity, for example, electrical polarity. He even gave the calculus of quaternions a "semi-metaphysical" interpretation as a "calculus of polarities", echoing his earlier interest in idealist philosophy [**17**, 2:436, 2:440]. In an even stronger metaphysical vein, he came to view quaternions as a natural algebra of space and time, in which the three dimensions of space were joined by the fourth dimension of time. "The quaternion (was) born, as a curious offspring of a quaternion of parents, say of geometry, algebra, metaphysics, and poetry", he wrote and quoted his own sonnet addressed to Sir John Herschel as the clearest expression of the quaternion idea:

> "And how the one of Time, of Space the Three,
> Might in the Chain of Symbols girdled be"

(quoted in [**20**, p. 176]).

Hamilton did everything he could to make his theory unpalatable to the reader, choosing from the outset a "*metaphysical style* of expression". Herschel implored him to make his ideas "clear and familiar down to the level of ordinary unmetaphysical apprehension" and to "introduce the new phrases as strong meat gradually given to babes" [**17**, 2:633], but to no avail. Hamilton loaded his 800-page-long *Lectures on Quaternions* (1853) with new impenetrable terminology, such as *vector, vehend, vection, vectum, revector, revehend, revection, revectum, provector, transvector, factor, profactor, versor,* and